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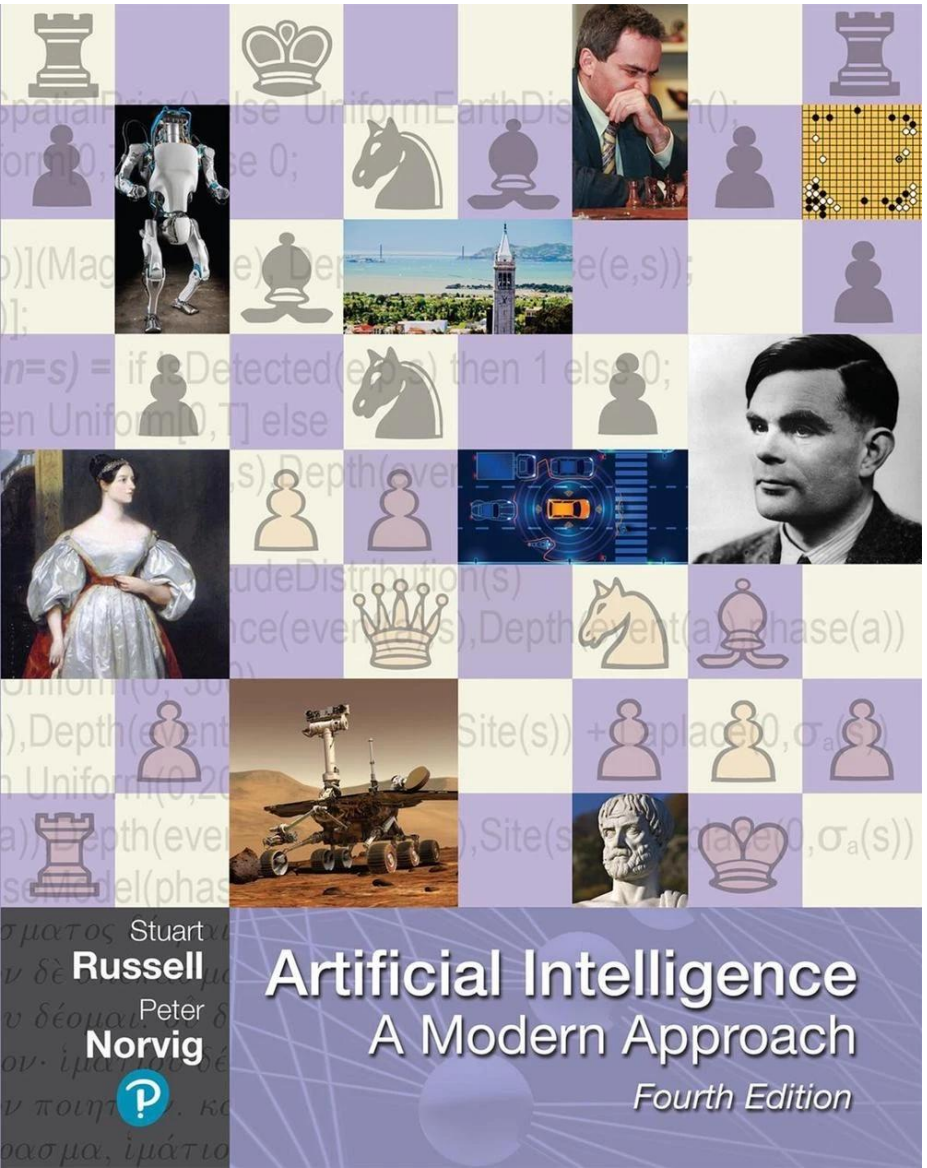

Chapter 24

Deep Learning for Natural Language Processing

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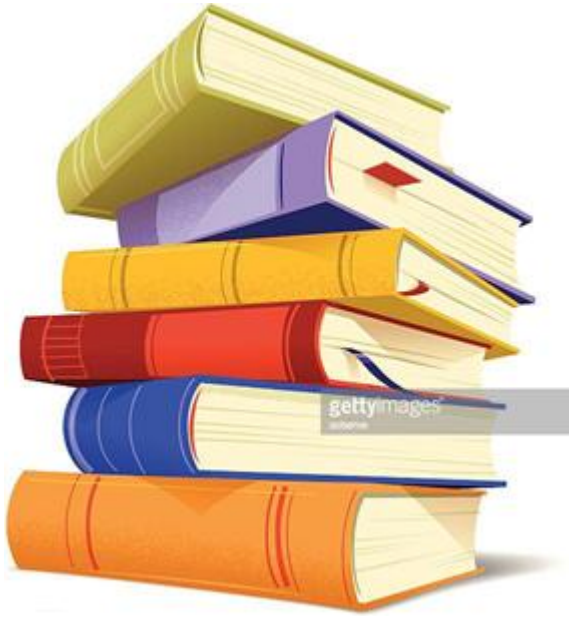
Contents



- Word Embeddings
- Recurrent Neural Networks for NLP
- Sequence-to-Sequence Models
- The Transformer Architecture
- Pretraining and Transfer Learning
- State of the art

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Word Embeddings

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Why Deep Learning Changed NLP

- Language has **structure**, but also endless variation.
- Neural models learn representations directly from data.
- Dense vectors support generalization across related words and contexts.
- Chapter 24 extends Chapter 23 from symbolic models to learned representations.

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From One-Hot Symbols to Embeddings

- **One-hot vectors** identify words but do not encode similarity.
- **Word embeddings** map words to dense, low-dimensional vectors.
- Related words tend to occupy nearby regions in vector space.
- Embeddings are learned automatically from text or task supervision.

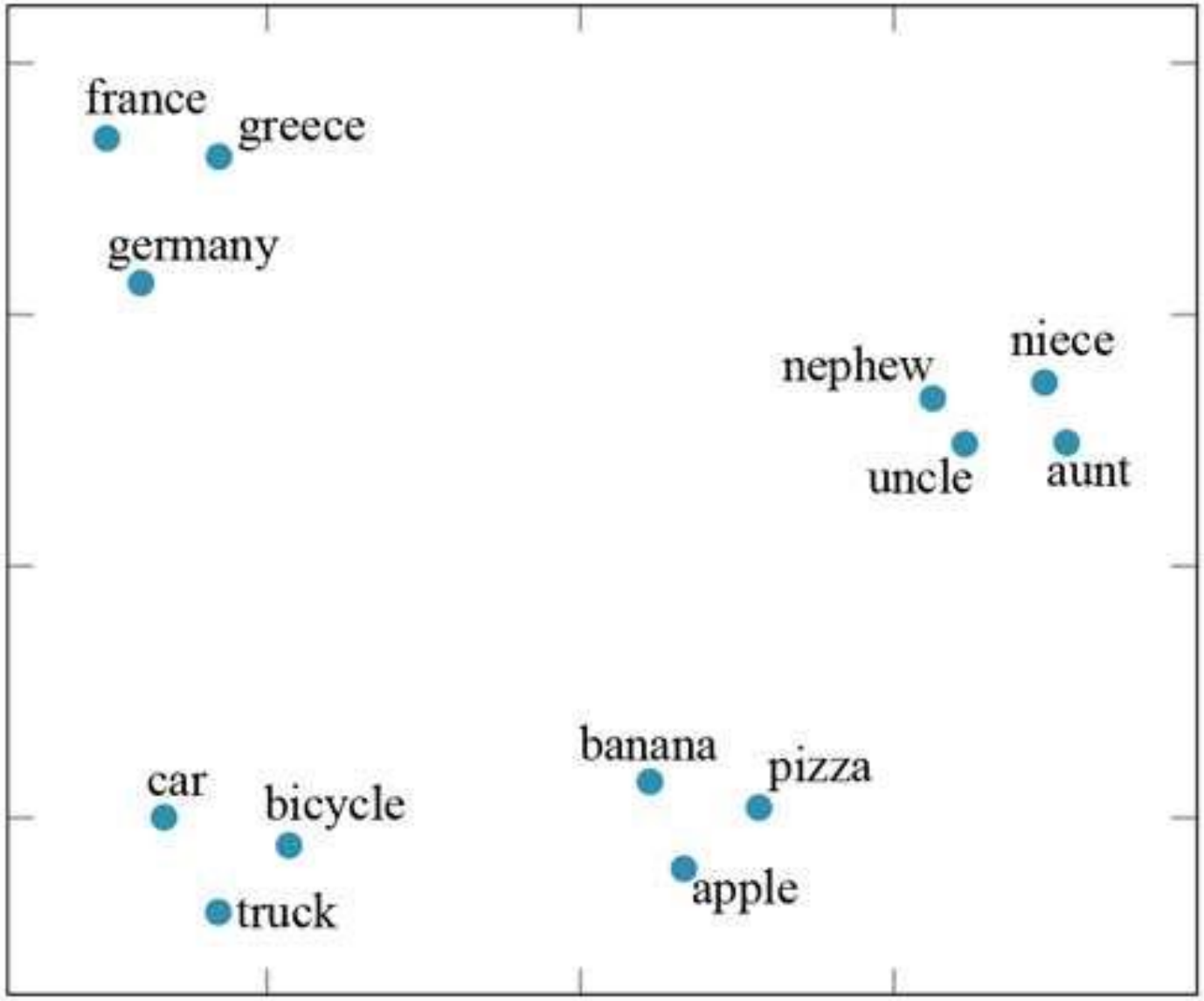

Figure 24.1. GloVe word embeddings projected into two dimensions.

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What Embedding Dimensions Mean

- Individual dimensions usually lack clear human-readable meanings.
- The **geometry** of the whole vector space is meaningful.
- Similarity is often measured by distance or cosine similarity.
- Embeddings support downstream tasks more than direct interpretation.

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Embedding Geometry and Analogies

- Vector offsets can capture regularities: **king − man + woman ≈ queen**.
- Analogies are solved by searching for the nearest vector.
- The effect is useful but not guaranteed for every relation.
- Quality depends on corpus, training objective, and vocabulary coverage.

A	B	C	D = C + (B − A)	Relationship
Athens	Greece	Oslo	Norway	*Capital*
Astana	Kazakhstan	Harare	Zimbabwe	*Capital*
Angola	kwanza	Iran	rial	*Currency*
copper	Cu	gold	Au	*Atomic Symbol*
Microsoft	Windows	Google	Android	*Operating System*
New York	New York Times	Baltimore	Baltimore Sun	*Newspaper*
Berlusconi	Silvio	Obama	Barack	*First name*
Switzerland	Swiss	Cambodia	Cambodian	*Nationality*
Einstein	scientist	Picasso	painter	*Occupation*
brother	sister	grandson	granddaughter	*Family Relation*
Chicago	Illinois	Stockton	California	*State*
possibly	impossibly	ethical	unethical	*Negative*
mouse	mice	dollar	dollars	*Plural*
easy	easiest	lucky	luckiest	*Superlative*
walking	walked	swimming	swam	*Past tense*

Figure 24.2. Word analogies solved with embedding-vector arithmetic.

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Supervised Embeddings for POS Tagging

- A neural tagger can learn embeddings while learning the task.
- A fixed-size **word window** provides local context.
- The middle word is classified using surrounding embeddings.
- Embedding weights and network weights are trained together.

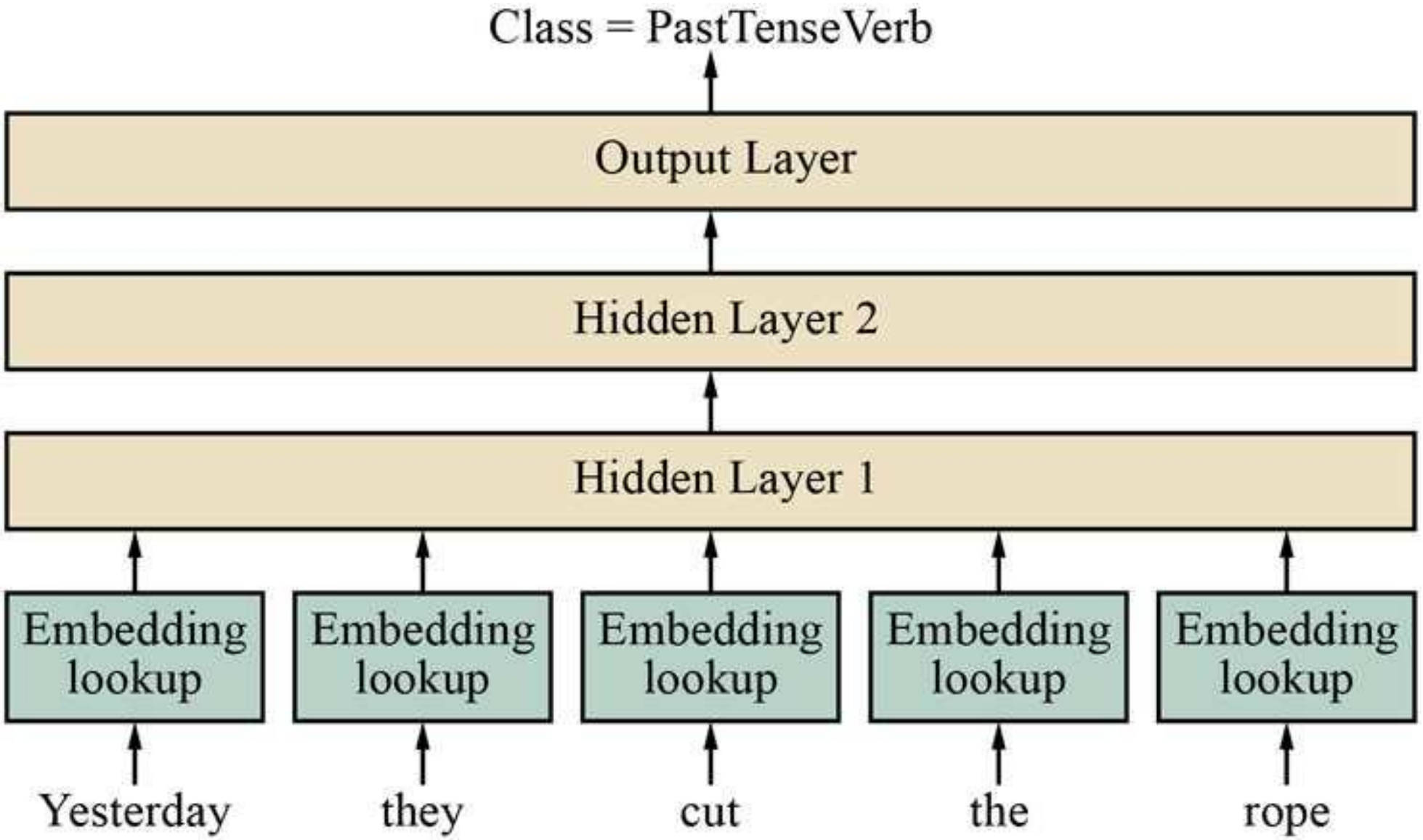

Figure 24.3. Feedforward POS tagging model using a five-word window.

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Character-Level Models

- Characters can be encoded as one-hot vectors too.
- Character models learn morphology from prefixes, suffixes, and spelling.
- They handle rare or unseen words better than fixed word vocabularies.
- Most NLP systems still use word or subword-level representations.

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Limits of Static Word Embeddings

- A word receives one vector regardless of context.
- Polysemy is hard: **rose** can be a flower or a verb form.
- Idioms may not equal the sum of their component words.
- Biases in training corpora can be reflected in embeddings.

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Recurrent Neural Networks for NLP

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Why RNNs for Language?

- Language is an **ordered sequence**, not a fixed bag of words.
- RNNs carry a hidden state from one timestep to the next.
- The hidden state can store relevant prior context.
- The same weights are reused across sequence positions.

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RNN Language Models

- Input at timestep t is the embedding of word w_t.
- The hidden state summarizes previous words.
- The output is a softmax over the vocabulary for w_{t+1}.
- Training uses unlabeled text: the next word is the label.

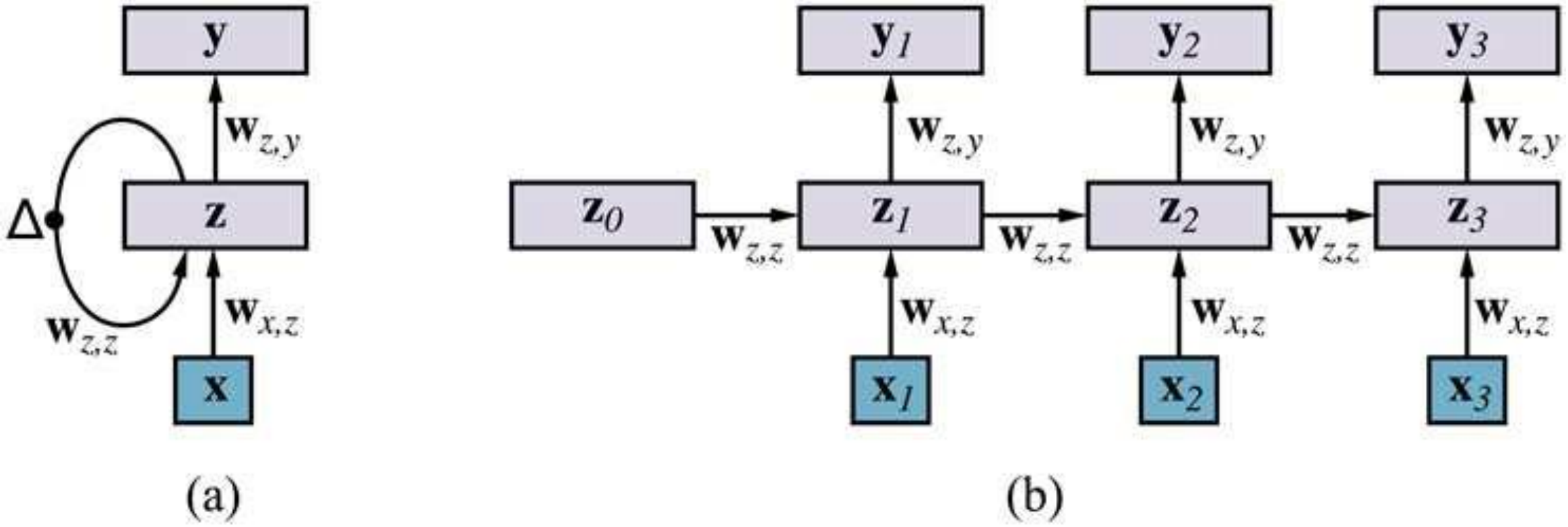

Figure 24.4. A recurrent neural network and its unrolled computation.

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Training an RNN Language Model

- Use text as a sequence of input-output pairs shifted by one word.
- For **hello world**, input hello predicts output world.
- Cross-entropy loss compares softmax output with the true next word.
- Back-propagation through time updates shared recurrent weights.

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RNNs for Token-Level Classification

- Each timestep can output a label for the current word.
- POS tagging predicts tags; coreference predicts antecedents.
- Training requires labeled sequences, not just raw text.
- Bidirectional RNNs use both left and right context.

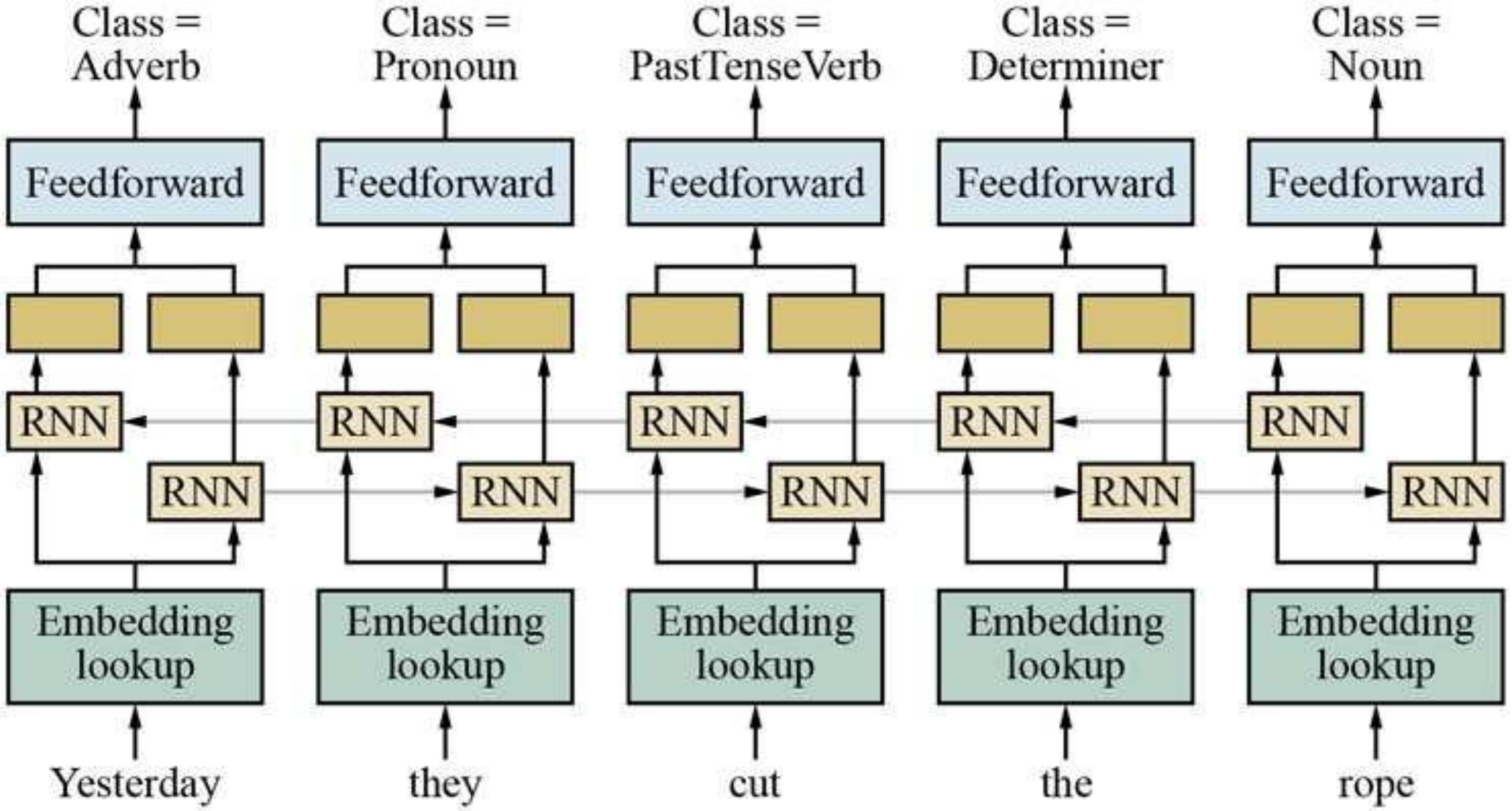

Figure 24.5. A bidirectional RNN network for POS tagging.

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Sentence-Level Classification

- Some tasks need one label for the whole sentence or document.
- Examples: **sentiment**, topic, intent, spam detection.
- The final hidden state can summarize the input sequence.
- Pooling over hidden states is another common strategy.

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LSTMs and the Long-Context Problem

- Plain RNNs can lose or distort information over many steps.
- This resembles the **vanishing gradient** problem.
- LSTMs add gates to control what to remember or forget.
- They improved many NLP tasks before transformers became dominant.

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RNN Strengths and Weaknesses

- Strength:
  - Naturally processes variable-length sequences.
  - Hidden states adapt to task-relevant information.
- Weakness:
  - Sequential computation limits parallelism.
  - Very long dependencies remain difficult.

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Sequence-to-Sequence Models

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Machine Translation as Sequence Transduction

- Input and output are both sequences, often with different lengths.
- The target word depends on the full source sentence and prior target words.
- Training uses aligned source-target sentence pairs.
- The goal is fluent text that preserves meaning.

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Encoder-Decoder Architecture

- The **encoder** reads the source sequence.
- The **decoder** generates the target sequence.
- A hidden representation connects source understanding to target generation.
- Generation stops when the model emits an end-of-sequence token.

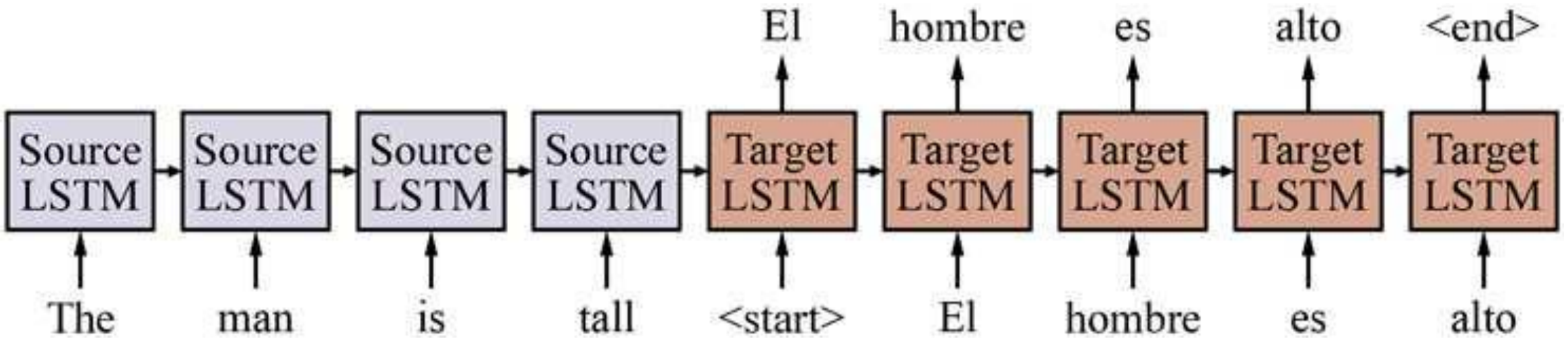

Figure 24.6. A basic sequence-to-sequence model for translation.

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Three Shortcomings of Basic Seq2Seq

- **Nearby context bias**: hidden state favors recent words.
- **Limited context**: source sentence is compressed too much.
- **Expensive search**: target sequences create many hypotheses.
- These limitations motivate attention and better decoding.

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Attention: The Core Idea

- The decoder should focus on different source words at each step.
- Attention assigns a **soft weight** to each source position.
- A context vector is the weighted sum of source hidden states.
- The mechanism is differentiable and trainable end-to-end.

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Attention Equations

- Score: $\boldsymbol{r_{ij} = h_{i-1}.s_j}$
- Normalize: $\boldsymbol{a_{ij} = exp(r_{ij})/\Sigma_k exp(r_{ik})}$
- Context: $\boldsymbol{c_i = \Sigma_j a_{ij} s_j}$
- Softmax turns raw scores into a probability distribution.

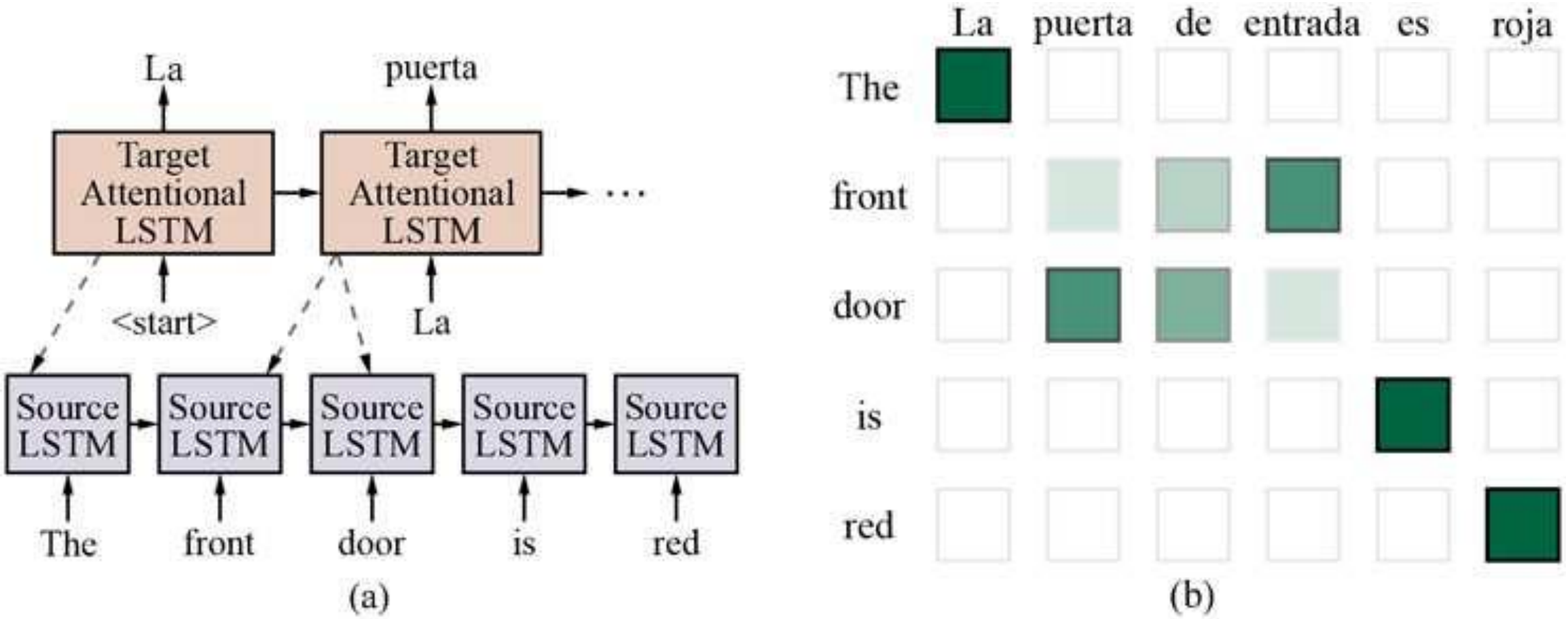

Figure 24.7. Attention links target words to relevant source words.

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Decoding: Greedy vs Beam Search

- **Greedy decoding** chooses the best next word at each step.
- It may miss the best whole sentence.
- **Beam search** keeps the top b partial hypotheses.
- Hypothesis score is the sum of log word probabilities.

Figure 24.8. Beam search keeps multiple translation hypotheses.

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Beam Size Tradeoff

- Small beams are faster but risk missing good translations.
- Large beams explore more hypotheses but cost more computation.
- Very large beams may favor overly short or generic outputs.
- Decoding is a search problem over possible target sequences.

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The Transformer Architecture

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Why Transformers?

- RNNs process sequences one step at a time.
- Transformers use **self-attention** to connect all positions directly.
- They model long-distance context without sequential dependency.
- Matrix operations make them efficient on GPUs and TPUs.

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Self-Attention

- Each word attends to other words in the same sequence.
- Self-attention creates a context-aware vector for each position.
- The relation is **asymmetric**: attention from i to j can differ from j to i.
- All positions can be processed in parallel.

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Queries, Keys, and Values

- Each input vector is projected into **query**, **key**, and **value** vectors.
- Query asks: what am I looking for?
- Key answers: what information do I contain?
- Value provides the content to be combined.

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Scaled Dot-Product Attention

- Raw score: dot product between query and key.
- Scale by $\sqrt{d}$ for numerical stability.
- Softmax converts scores into attention weights.
- Output is the weighted sum of value vectors.

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From Self-Attention to Transformer Layers

- A transformer layer includes self-attention and a feedforward network.
- **Residual connections** help gradients flow through deep stacks.
- Layer outputs become inputs to the next layer.
- Practical models usually stack many layers.

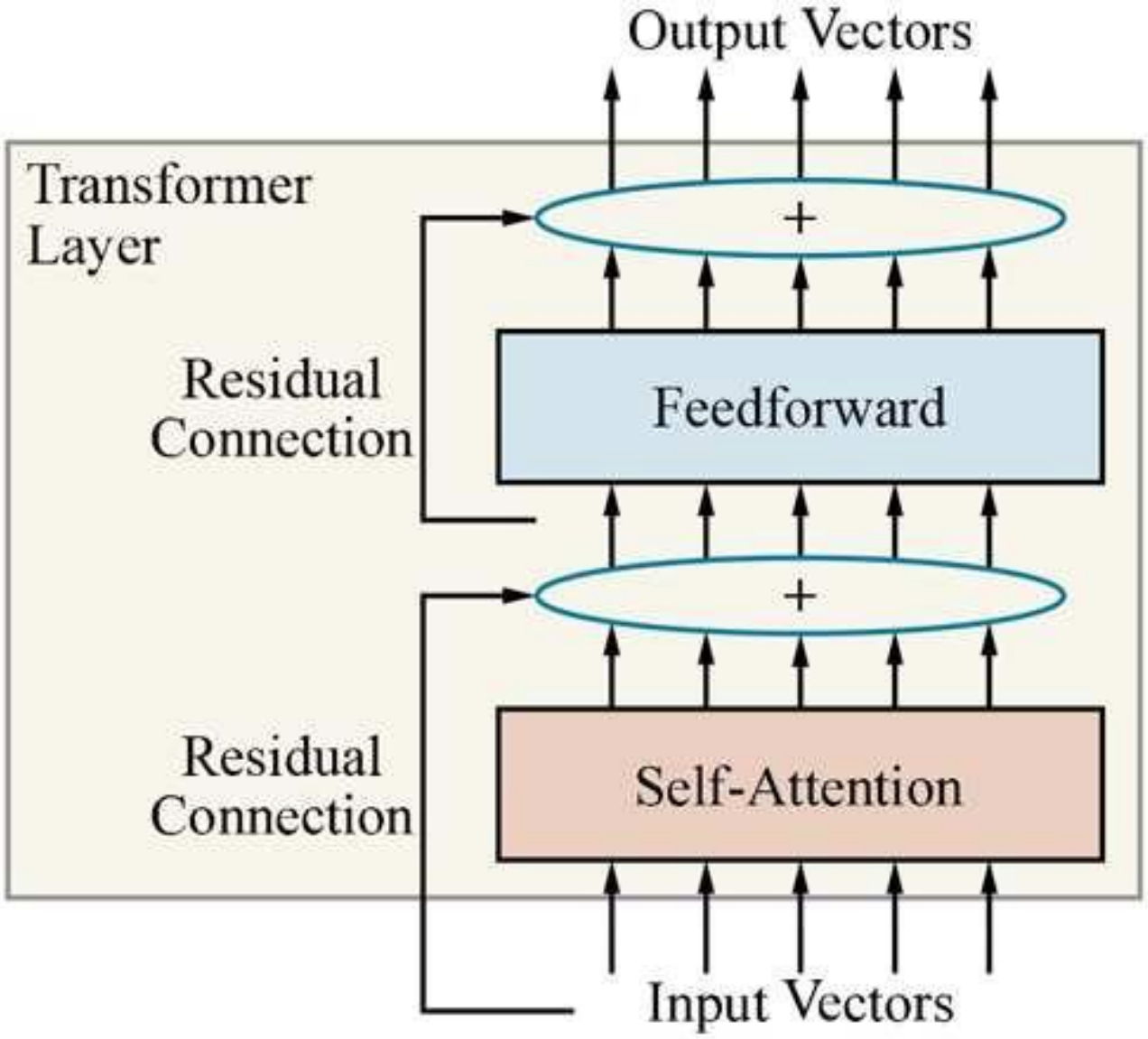

Figure 24.9. A single-layer transformer with self-attention and feedforward sublayers.

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Positional Embeddings

- Self-attention alone is insensitive to word order.
- Transformers add **positional embeddings** to word embeddings.
- The input vector represents both word identity and position.
- Order information becomes available to every layer.

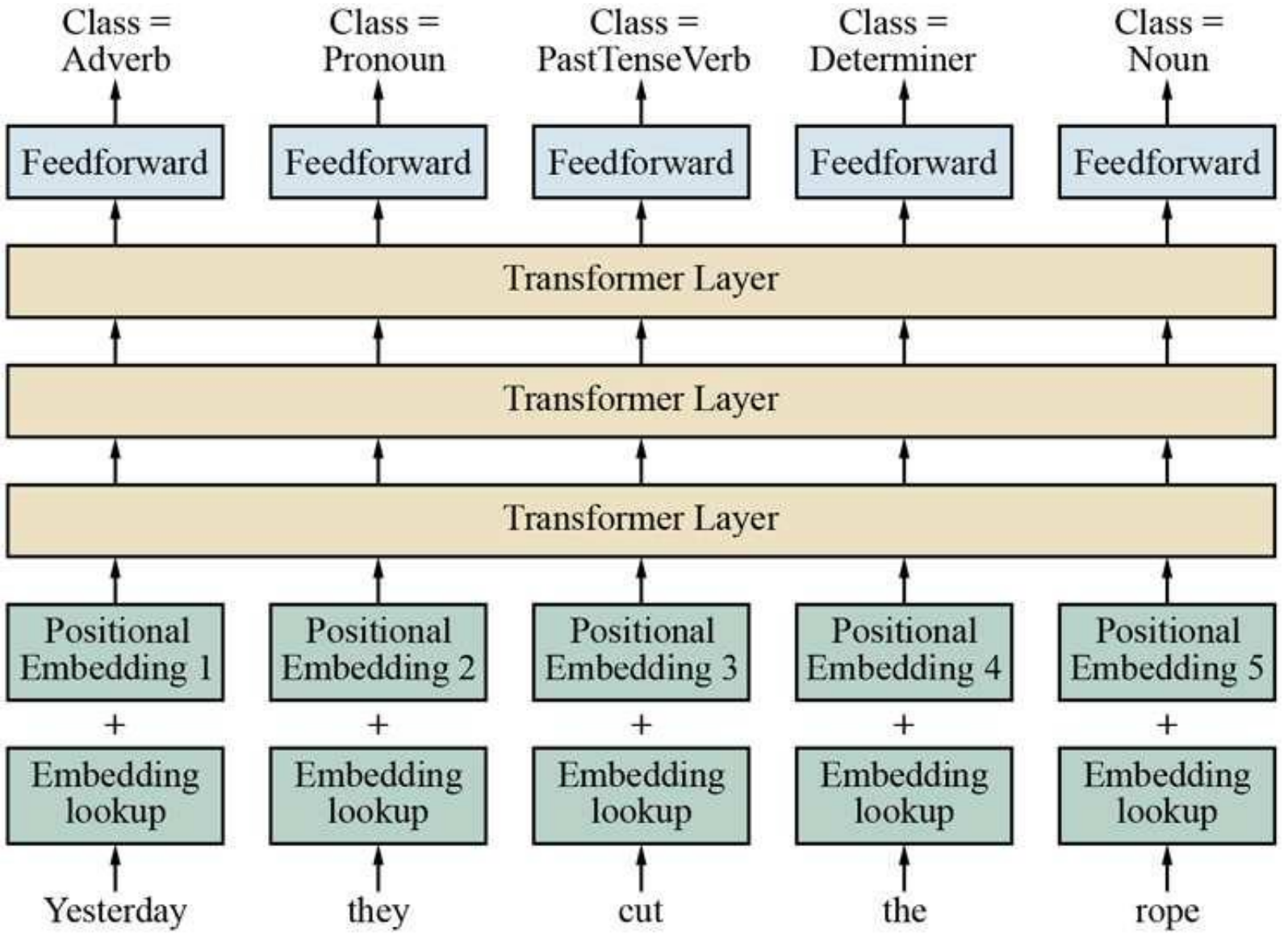

Figure 24.10. Transformer architecture for POS tagging.

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Transformer Encoder and Decoder

- The chapter first describes the **transformer encoder**.
- Encoders are useful for classification and tagging tasks.
- Full machine translation also uses a **transformer decoder**.
- The decoder attends to encoder outputs while generating target text.

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Transformers vs RNNs

- RNNs encode order naturally but compute sequentially.
- Transformers need position information but compute in parallel.
- Self-attention gives shorter paths between distant words.
- The cost grows with sequence length because all pairs are considered.

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Pretraining and Transfer Learning

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Why Pretraining Fits NLP

- Labeled linguistic data is expensive to create.
- Unlabeled text is abundant on the Internet.
- Pretraining learns general linguistic knowledge from raw text.
- Fine-tuning adapts the model to a specific task.

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Pretrained Word Embeddings

- WORD2VEC and GloVe learn vectors from large unlabeled corpora.
- GloVe uses word **co-occurrence counts** within a context window.
- Dot products approximate log co-occurrence probabilities.
- Pretrained vectors can be reused in downstream models.

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Domain-Specific Embeddings

- Embeddings can encode knowledge from specialized corpora.
- Scientific, medical, or legal text contains domain-specific relations.
- A general embedding may miss these relations.
- Training on the right corpus can improve downstream performance.

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Contextual Representations

- A contextual vector depends on both the word and its surrounding context.
- The same word can receive different vectors in different sentences.
- This addresses **polysemy** and some idiomatic usage.
- Contextual models became central to modern transfer learning.

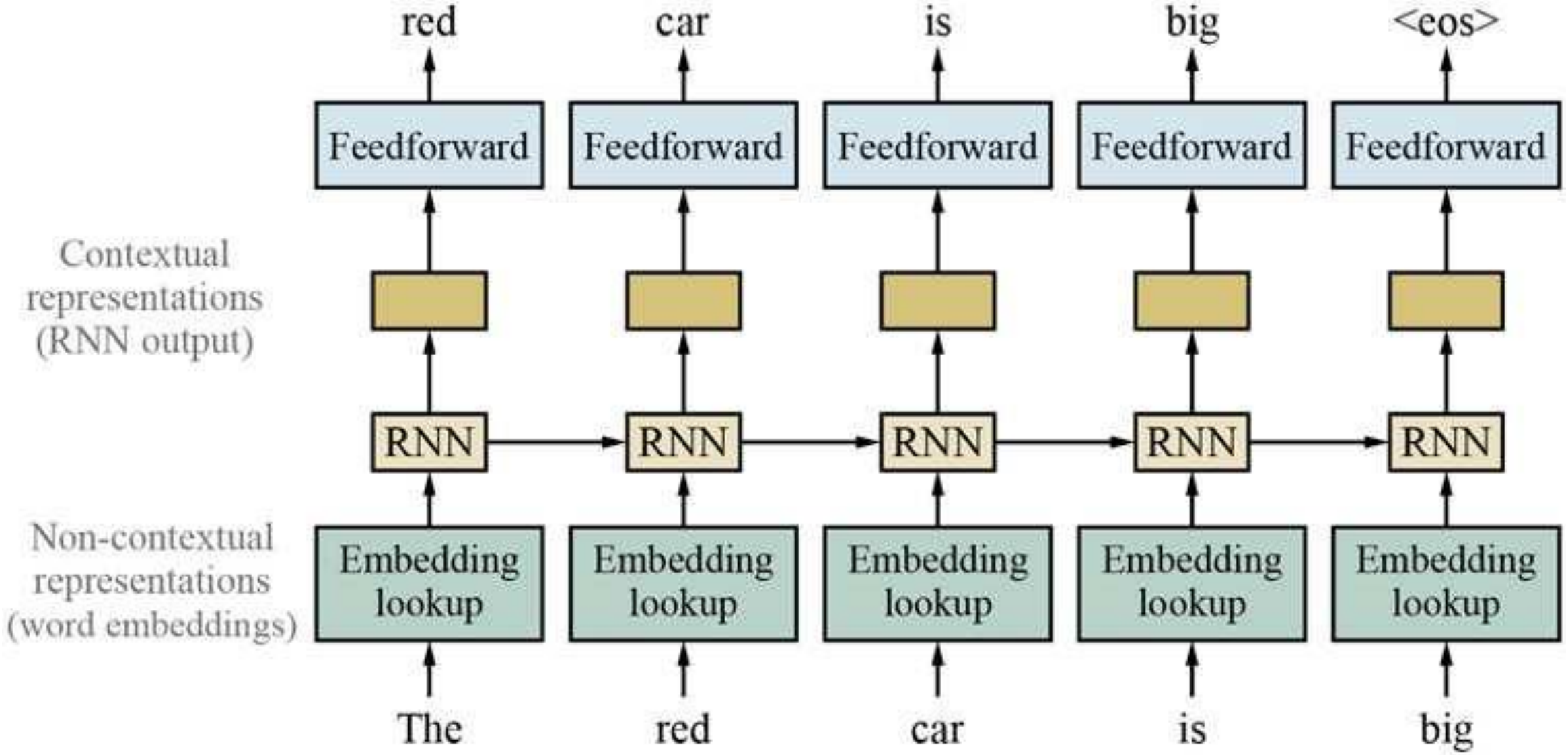

Figure 24.11. Training contextual representations with a left-to-right language model.

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Masked Language Models

- Mask some input words and train the model to recover them.
- The sentence itself supplies the training label.
- Bidirectional context can be used around the mask.
- After pretraining, the model can be fine-tuned for many tasks.

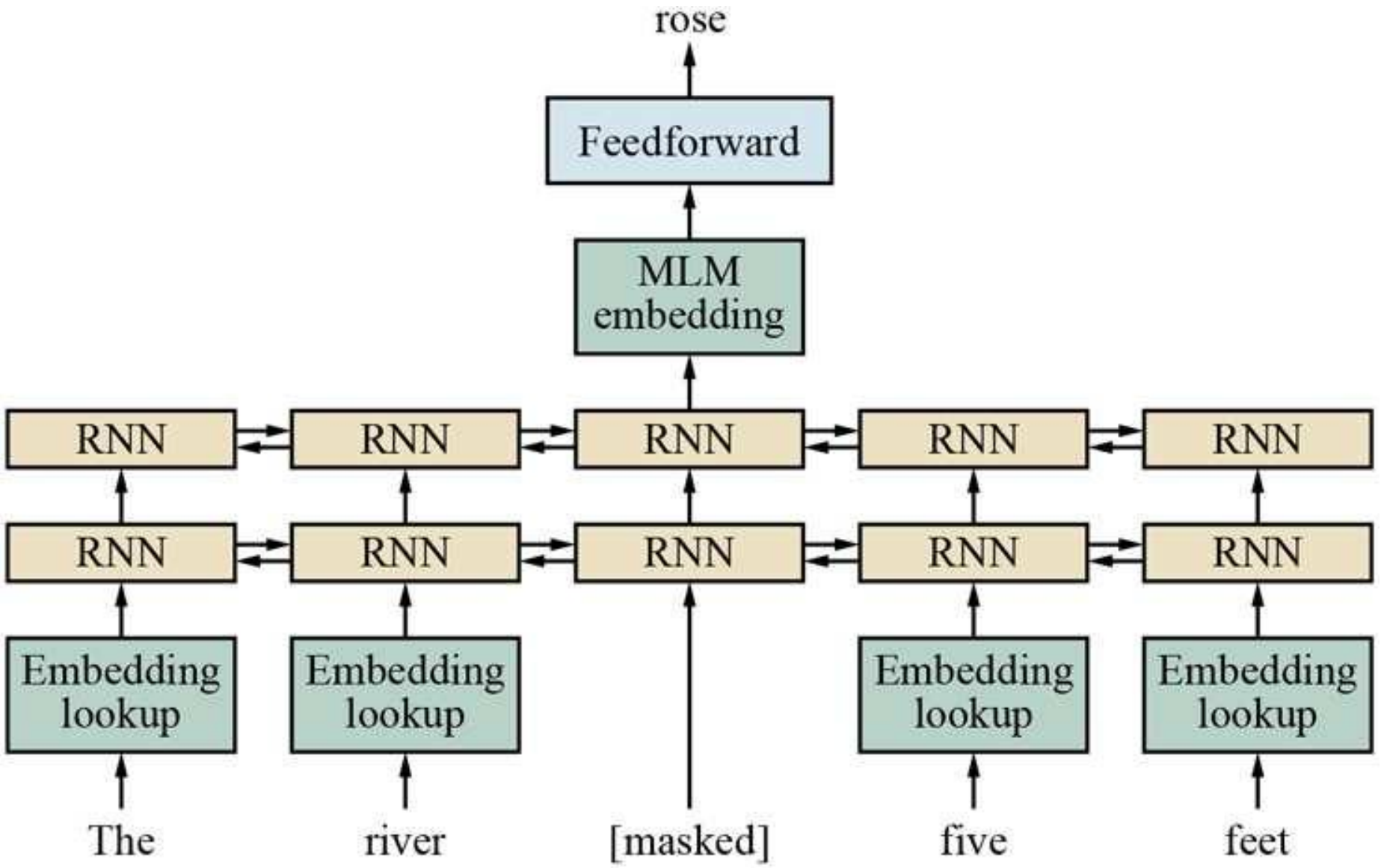

Figure 24.12. Masked language modeling predicts masked input words.

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Fine-Tuning and Transfer

- Start with a model pretrained on massive unlabeled text.
- Add or adapt an output layer for the target task.
- Train further on a smaller labeled dataset.
- The same pretrained model can support QA, entailment, tagging, and classification.

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State of the art

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NLP's ImageNet Moment

- Around 2018, transfer learning produced rapid NLP progress.
- Pretrained models became reusable foundations for many tasks.
- Hardware advances made very large models feasible.
- Downloading a pretrained model became more practical than training from scratch.

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Milestones in Modern NLP

- **WORD2VEC** and **GloVe** popularized reusable word vectors.
- **ELMo** introduced deep contextual word representations.
- **BERT** used masked language modeling for bidirectional pretraining.
- **GPT-style models** showed strong left-to-right generation.

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ARISTO and Question Answering

- ARISTO answered 8th-grade science multiple-choice questions.
- It used an **ensemble** of solvers, including large language models.
- ROBERTA alone achieved a strong score on the exam.
- The task requires language, commonsense, and science knowledge.

1. **What will best separate a mixture of iron filings and black pepper?**
   (a) magnet (b) filter paper (c) triple beam balance (d) voltmeter
2. **Which form of energy is produced when a rubber band vibrates?**
   (a) chemical (b) light (c) electrical (d) sound
3. **Because copper is a metal, it is**
   (a) liquid at room temperature (b) nonreactive with other substances
   (c) a poor conductor of electricity (d) a good conductor of heat
4. **Which process in an apple tree primarily results from cell division?**
   (a) growth (b) photosynthesis (c) gas exchange (d) waste removal

Figure 24.13. ARISTO answers 8th-grade science questions.

It is a truth universally acknowledged that the earth is not the center of the universe. There are those who assert there is. I do not accept them, but others I consider to be of the same opinion. The truth is, however, that if there are other than the center, and if there are any other living things in the universe and if they are not human, then we do not yet have our answers. We have to go on. This page gives a simplified, simplified answer to the problem. We don't have all the answers. The truth is, however, that the truth is out there.

When Gregor Samsa woke up one morning, he did not notice anything strange. "When my wife is looking at me, I feel like she is looking at a piece of art," he said. "I think she is admiring something I have created." The idea is that by looking at your own life, you learn something important and become a better person. It is a theory that emerged from psychologist Daniel Goleman's work, in which he asked "How do you know you're not a loser?"

Alice was beginning to get very tired of sitting with her sister on the bank. She sat up, yawned, and said, with a loud little scream, "I hope you don't mind if I keep on doing what I should like to do, and if someone asks me which of us will do more, don't tell them that I won't do much, my dear sister."

All happy families are alike; each happy family is like a garden of paradise. The only difference between happy families and unhappy families, is that the unhappy family doesn't have any flowers or trees.

Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Tell me a story. Please fill out the following details. Thank you... Thank you for your interest in this interview. Please wait...

Figure 24.14. GPT-2 completions show fluent local generation and occasional failure.

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Current Limitations

- Large models require enormous text and compute.
- Transformer context is often limited to a few hundred or thousand tokens.
- Models can produce fluent but incorrect or unstable text.
- They still lag humans on many forms of robust understanding.

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Toward Hybrid Approaches

- Data-driven models are easier to build and maintain today.
- They may learn latent structure similar to grammar and semantics.
- But we do not fully understand what large models represent.
- Future systems may combine neural learning with explicit structure.

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Summary

- **Word embeddings** replace atomic symbols with robust continuous vectors.
- **RNNs and seq2seq models** handle ordered text and generation.
- **Attention and transformers** model long-distance context efficiently.
- **Pretraining and transfer learning** turn unlabeled text into reusable knowledge.
- **State-of-the-art NLP** is powerful, but still limited and data-hungry.


Q
A